1.66 Forces: Equilibrium

Worked Solutions

1 a)

15 N upwards

1 b)

7 N left

2 a)

$P = 2$

2 b)

$P = 6$

2 c)

$P + 4 = 38 \Rightarrow P = 34$

2 d)

$40 + P = 3P \Rightarrow P = 20$

3 a)

$P = 62$ and $Q = 31$

3 b)

$16 + P - 3 = 40$ so $P = 27$

$12 + 4 = 2Q$ so $Q = 8$

3 c)

$P = 12$ so $3Q = 12 \Rightarrow Q = 4$

4 a)

$P = 2Q + 4$, $10 = P + Q$ So $P = 8$, $Q = 2$

4 b)

$7 = P + 2Q$, $12 = 2P + 3Q$ $P = 3$, $Q = 2$

5 a)

$T = 50$N

5 b)

$50 = 10m$ so $m = 5$ kg

5 c)

50 N downwards

6 a)

$w = 50 \times 9.81 = 490.5$ N

491 N (3 SF – AQA)

6 b)

$w = 50 \times 1.63 = 81.5$ N

7)

$4000 \div 9.8 \times 1.6 = 653$ N (3 SF)

650 N (2 SF – AQA)

8 a)

$0.4 \times 9.8 = 3.92$ N

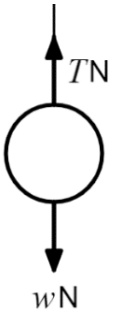
8 b)

$P = 3.92$

8 c)

3.92 N downwards

9 a)



9 b)

$2 \times 9.8 = 19.6$ N

9 c)

$34.3 \div 9.8 = 3.5$ kg

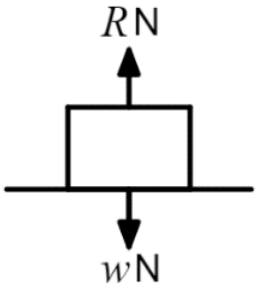
9 d)

$0.5 \times 9.8 = 4.9$ N

9 e)

14.7 N

10 a)



10 b)

$1.5 \times 10 = 15$ N

10 c)

$60 \div 10 = 6$ kg

10 d)

$5 \div 10 = 0.5$ kg

10 e)

$1.2 \times 10 = 12$ N

10 f)

25 N

11)

